

Jargon Buster



Beats: The periodic variations produced in the loudness of a sound. Beats can be heard when two tones of slightly different frequencies are produced at the same time.

Crossover: Most speakers consist of just one full-range speaker cone (driver) that is used to generate all the audio frequencies. However, in speakers that use a Woofer, Squaker and Tweeter, the audio frequencies need to be split and delivered to these specific drivers. The crossover is a circuitry that is integrated into speakers. It splits the audio frequencies before delivering them to the specific drivers—low frequencies are sent to the Woofer, mid-frequencies to the Squaker and high frequencies to the Tweeter.

Decibels: A logarithmic scale of measurement, the decibel is used to indicate the relative loudness of sound.

Dynamic range: The range exhibited by a speaker system in producing very soft and very loud sounds without distortion.

Frequency: It is the number of complete sound waves that pass a point each second and is measured in Hertz (Hz). To put things into perspective, an Audio CD has a maximum output frequency of 20 KHz, which means that it can effectively playback the entire spectrum of sounds audible by humans (the ear can detect frequencies between 20 Hz to 20 KHz).

Frequency response: This is the ability of the speaker system to respond faithfully to a range of frequencies that it is fed with across a frequency spectrum. Ideally, a speaker system should

have a flat frequency response for all frequencies between 20 Hz and 20 KHz as this is the audio range that humans can hear.

LFE: Often confused with the subwoofer channel, LFE (Low Frequency Effects) is a channel of low-frequency audio information in surround sound standards such as Dolby Digital. It contains low frequencies as experienced in sound effects such as explosions, gunshots, thunder, etc. This channel is discrete in the Dolby Digital standard and is the '1' in the 5.1 naming convention. In the absence of a dedicated subwoofer in your home theatre system, the LFE channel would need to be distributed over the other channels. The subwoofer is simply a speaker that is tailored to handle low frequencies and it is therefore paired with the LFE channel in a surround sound system.

Power rating: This is a specification of the maximum power that a speaker utilises during operation. Obviously, the greater this value, the louder the speaker will sound.

Woofers, Tweeters and Squakers: The names of speaker drivers (cones) that are built to reproduce specific frequency ranges. Woofers are low-frequency drivers, Tweeters are high-frequency drivers, while Squakers are mid-range drivers. Speaker systems are generally based on a Woofer-Tweeter or a Woofer-Squaker-Tweeter combination.

egory, from speakers that have a cool and funky design to the purely functional ones without any of the frills and embellishments. The only 2.1 speaker that offered volume control in the form of a convenient in-line controller was the Creative Inspire 2.1 2400. The Sumoku Alpha - 4R was one of the few speakers that came with an external amplifier that houses the volume, bass, treble, mute and the power controls. In fact, even the other two 2.1 speaker systems from Sumoku (Alpha - 5 and Alpha - 6) featured these controls—a definite plus point. Having an external volume control eliminates the need for accessing the volume control in Windows, which can be quite cumbersome. It is also far more convenient to have the volume control on a separate amplifier than on the subwoofer as it gives you the freedom to place the subwoofer anywhere since you don't have to access it. The Sumoku speakers were also the most complete speaker systems in terms of the details and thoroughness of the manuals and documentation.

The Mercury SW480 featured one of the smallest subwoofers in the 2.1 speaker category. It is usually easy to set up a 2.1 speaker system, but this one featured extremely short cables, making it rather difficult to set it up on a table, especially if you have a large monitor and need to place the speakers apart. This was also apparent with the subwoofer—since they are usually placed on the ground, there was a bit of a problem due to the short cable.

A four-channel speaker system is ideally paired with a four-channel soundcard and it would make sense to go in for one of these if you intend to buy such a soundcard in the near future. You can even opt to upgrade to a 4.1 setup if you already possess a four-channel soundcard but are stuck with your old two-speaker setup. Though referred to as 4.1 speaker systems, it is implied that these speakers have discrete Line In connections for the front and the rear stereo channels. The '1' channel is the subwoofer that is derived from the front channels and is not a separate channel.

Interestingly, even though we received some speakers branded as 4.1 surround speakers, they were in fact pseudo 4.1 systems. In reality, they were 2.1 speaker systems that featured a single stereo input that is taken from the Line Out port of your soundcard—the rear inputs were missing altogether. Therefore, even though they have four separate satellite speakers, the rear speakers receive the exact same signal as the front speakers!

The one that stood out in the 4.1 speaker category was the Labtec Arena 530. This four-channel speaker system featured satellites that looked like something straight out of a *Star Trek* studio set—they had an eye-catching design, with curvy, well-moulded grilles and speaker enclosures. Apart from the main volume and the rear volume buttons on the subwoofer, this one also featured the power button and a special button called M3D (Matrix 3D), which helps you simulate four channels from a stereo source.

One of the most feature-rich 4.1 speakers was the Logitech Xtrusio DSR 100. It features discrete inputs for a 4.1 setup and also sports an environmental equalizer knob, which lets you vary the crossover frequency of the subwoofer across two frequencies, namely 100 Hz and 10 KHz. The shape of the subwoofer is far from conventional and the black colour finish and its oval design makes it look striking. This speaker system also features a digital connector in addition to the conventional analogue inputs.

The Creative Inspire 4.1 4400 also provides a very convenient in-line remote control for control-



Logitech Xtrusio DSR 100: A feature-rich 4.1 speaker set